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Shipping live sea urchins: effects of temperature and exposure time on survival

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Abstract

Sea urchin aquaculture could offset declines in wild fisheries. The economic feasibility of sea urchin aquaculture may depend on farmers receiving premium prices for product transported live internationally to markets or processing facilities. Research into cost-effective methods of transporting live sea urchins out of water is scarce. In this study, to get a baseline of the temperatures experienced by sea urchins transported by air in the Indo-Pacific region, temperature loggers recorded temperatures in ‘cooler boxes’ containing the sea urchin, *Tripneustes gratilla*, packed and shipped in seawater or dry. We then examined the effect of time kept out of water (6, 12, 24, 48 h) and air temperature (chilled ~4 °C, cooled ~12 °C, or ambient ~27 °C) on mortality, change in wet weight, and fitness (as measured using righting behaviour) of two sea urchins, *T. gratilla* and *Heliocidaris erythrogramma*. For both species mortality (15 days after the treatment) was low (0–20%) when held out of water for up to 24 h at 12 °C, but 100% died within 15 days when held out of water at 4 °C and 27 °C. Mortality increased (~60%) for *T. gratilla* held out of water for 48 h at 12 °C, but remained at ~20% for *H. erythrogramma* in the same treatment. Righting times were faster for urchins exposed to air at 12 °C than at 4 or 27 °C. The mass (fluid) lost by urchins was a key predictor of mortality with urchins losing a higher percentage of their mass during longer exposure periods at higher temperatures. *T. gratilla* frequently spawned after being out of water. To test packing strategies that could influence sea urchin health, *T. gratilla* were packed out of water at 28.6 °C for 12 h in sealed or open containers and in containers that drained off excreted fluid or retained fluid around the urchins. None of these treatments had an impact on mortality to 15 days, the amount of fluid lost, or righting activity. Our results indicate producers may be able to transport *T. gratilla* and *H. erythrogramma* dry for up to 24 h with little mortality as long as temperatures are maintained at ~12 °C. Additional research into limiting the fluid loss of sea urchins while being shipped dry may improve survival during longer shipping periods.

1. Introduction

Sea urchins are a valuable seafood. Global sea urchin production is declining from a peak of 108–120,000 t in 1995 as wild populations are overfished (Andrew et al. 2002; Sun and Chiang, 2015). Aquaculture is seen as a way to meet growing demand, but is a nascent industry (Sun and Chiang, 2015). One impediment to the growth of sea urchin aquaculture is uncertainty about the best way to transport sea urchins to market (James and Evensen, 2022).

The Japanese market imported ~10,700 t of processed and live sea urchin products in 2019/21, equivalent to ~75% of annual global exports (FAO, 2020). Live sea urchins imports to Japan sell for as much as three times the price of sea urchins harvested and exported as fresh or frozen uni (roe) (Sun and Chiang, 2015). Live *Strongylocentrotus droebachiensis* are air freighted to Japan dry (out of water) in polystyrene foam boxes with frozen gel packs to maintain a low temperature, though specific information on packing protocols is not publicly available (James and Evensen, 2019). Although data for most species is unavailable, dry transport appears to be suitable if the duration is short (i.e., hours to days) but is not used when logistics chains are long because of high mortality rates (James and Evensen, 2022; Stefánsson et al., 2017; Sun and Chiang, 2015). Identifying reliable and cost-effective ways to transport sea urchins via complex logistics chains might enable more producers to access the lucrative live market when buyers or processors are distant from production centres.

The primary cause of mortality of sea urchins transported dry is thought to be desiccation (Dale et al., 2005). Desiccation may be exacerbated by the propensity of sea urchins exposed to air to spawn and excrete fluids from their oesophagus which can be equivalent to a loss of one-third of their body volume (Arafa et al., 2007; Burnett et al., 2002; James and Evensen, 2022). In a commercial trial, however, green sea urchins *S. droebachiensis* transported dry from Norway to Japan lost only 8% of their body weight (James and Evensen, 2018). It is possible that a small loss of fluid may have a positive effect by increasing humidity, as seen for crustaceans and mussels transported dry (e.g., Delorme et al., 2021; Samet and Nakamura, 1997). Sea urchins which replace internal fluid with air may also be better able to undertake gas exchange, thereby avoiding hypoxia (Burnett et al., 2002). A better understanding of how much fluid sea urchins lose while exposed to air and how fluid loss relates to mortality may give new insights into ways to increase the length of time that sea urchins can be transported dry.

One way to reduce mortality when transporting sea urchins dry may be to reduce the temperature. Lower temperatures generally improve the health of marine invertebrates exposed to air because respiration and metabolism are suppressed, which delays hypoxia and the negative consequences of anaerobic metabolism (e.g., de Andrade et al., 2021; Giomi et al., 2008; Morris and Oliver, 1999). The three studies that have tested the effects of temperature on sea urchins exposed to air suggest that lowering the temperature generally reduces mortality and increases the duration that sea urchins can cope with exposure, as long as temperatures are not below 0 °C (Arafa et al., 2007; James and Evenson, 2022; Miller and Bishop, 1973). For instance, the Mediterranean sea urchin *Paracentrotus lividus* experienced 100% mortality within 96 h after they were exposed to air at 25 °C for 5 h, but mortality was 40% when the air temperature was 19 °C (Arafa et al., 2007). Optimising temperatures during dry transport may also have the added benefit of improving the quality and shelf-life of gonads after harvest (Verachia et al., 2013). Because of the paucity of studies examining the impacts of temperature during dry transport of sea urchins, it is unclear whether optimal temperatures are species-specific or related to broad ecological characteristics (e.g., temperate vs tropical, subtidal vs intertidal).

This study investigated the effects of air temperature and duration of air exposure on the collector urchin, *Tripneustes gratilla* from the tropical Indo-Pacific, and the short-spined urchin, *Heliocidaris erythrogramma* from temperate Australia. Research on the aquaculture of *T. gratilla* has closed the lifecycle, refined culture techniques, and developed artificial diets to enhance gonad production and growth (e.g., Cyrus et al., 2015; Mos and Dworjanyn, 2019; Mos et al., 2011). Production runs of *T. gratilla* have been trialled in Australia, Japan, the Philippines, and South Africa for aquaculture (Cyrus et al., 2015; Juinio-Menez et al., 2008; Shimabukuro, 1991; pers. obs.) and in Hawaii for biocontrol of invasive seaweeds (Westbrook et al., 2015). Research on the aquaculture of *H. erythrogramma* has focused on improving gonad quality of wild-caught stock fed formulated diets in land-based systems (e.g., Pert et al., 2018; Warren-Myers et al., 2020; Warren-Myers et al., 2021), though the lifecycle has been closed and a chemical cue which induces settlement of larvae has been identified (Swanson et al., 2012). Little is known about the robustness of these two species when transported dry, but it may differ because *H. erythrogramma* can inhabit the high subtidal/low intertidal zone where they can be exposed to air, whereas *T. gratilla* occurs only in subtidal habitats (Lawrence and Agatsuma, 2013; Keesing, 2013).

To get a baseline of the temperatures experienced by sea urchins transported in the Indo-Pacific region, temperature loggers were used to record temperatures experienced by *T. gratilla* packed using wet (in seawater) and dry (without seawater) transport methods. To investigate how air temperature and the duration of air exposure may interact to influence robustness, *T. gratilla* and *H. erythrogramma* were held out of water in all combinations of four duration of air exposure treatments (6, 12, 24, or 48 h) and three air temperature treatments ('ambient' ~27 °C; 'cooled' ~12 °C; 'chilled' ~4 °C). Previous studies investigating the tolerance of sea urchins to dry transport measured mortality and effects on gonad index (Arafa et al., 2007; Arafa et al., 2010; Dale et al., 2005; James and Evensen, 2022; Kramer and Nordin, 1979; Warren and Pearce, 2020; but see Miller and Bishop, 1973). In this study, as well as measuring mortality to 15 days post-exposure, we monitored the amount of weight lost by sea urchins while out of water and investigated the sublethal effects of aerial exposure using an established protocol for monitoring the fitness of sea urchins, righting activity, following exposure to stress (Brothers and McClintock, 2015). We also tested whether mortality to 15 days, change in wet weight, and righting activity of *T. gratilla* were affected by the way in which they were packed. For this experiment, urchins were removed from water for 12 h in an air temperature of ~28 °C in containers with a sealed lid (sealed) or without a lid (open) where any fluid released by the sea urchins was drained away (drained) or allowed to pool around the bottom of the sea urchins' tests (not drained).

2. Material and Methods

2.1. Temperatures during wet and dry live freight

To document temperatures experienced by sea urchins when transported using wet (in seawater) and dry (out of water) methods in the Indo-Pacific, two temperature loggers (Thermodata IBCod G, set to record every 5 min) were placed in separate polystyrene foam boxes used to transport farmed *T. gratilla* from Coffs Harbour, New South Wales, Australia to Okinawa Island, Japan via air and road freight during 27–30 July 2017. *T. gratilla* were packed following conventional wet methods for transporting live marine animals; submerged in seawater inside plastic fish transport bags, pure O_2 gas used to fill the headspace, then housed within 20-L polystyrene foam boxes. Temperature loggers were added directly to the seawater to measure temperatures at the scale experienced by the sea urchins.

A temperature logger (Thermodata IBCod 22L, set to record every 5 min) was also placed in a package containing *T. gratilla* packed without seawater (i.e., dry) in sealed plastic fish transport bags housed in a 14-L polystyrene foam box and transported via road and air freight from Coffs Harbour, New South Wales, Australia to Adelaide, South Australia, Australia (21–22 October 2022).

2.2. Study species

For experiments investigating the effects of duration of air exposure, air temperature, and packaging, adult *T. gratilla* were collected near Cairns, QLD, Australia in April and June 2022 (Great Barrier Reef Marine Park Authority, collection permits G20/43835.1 and G22/47141.1) and transported in seawater to the National Marine Science Centre, Southern Cross University, Coffs Harbour (30°12.5'S, 153°16.1'E). *T. gratilla* were housed in six 60-L rectangular plastic tubs (~10 urchins per tub), supplied with heated (~27 °C) flow-through seawater at a rate of 59 L h^{-1} for at least four weeks before experiments began. Adult *H. erythrogramma* were collected near Coffs Harbour, NSW, Australia in April and May 2022 (NSW Department of Primary Industries, Scientific Collection Permit, P14/0014-2.0) and held in 200-L raceways supplied with ambient (~20 °C) flow-through seawater at a rate of 220 L h^{-1} for at least four weeks. Sea urchins were fed seaweed *Sargassum* spp. *ad libitum* until three days before experiments began. All sea urchins used for experiments could right themselves within 30 seconds and so were considered healthy.

2.3. Effects of air temperature and duration of air exposure on mortality, wet weight, and righting activity

Separate experiments for *T. gratilla* and *H. erythrogramma* tested the effects of four durations of exposure in air (6, 12, 24, 48 h) and three air temperatures (‘ambient’ ~27 °C; ‘cooled’ ~12 °C;

'chilled' ~4 °C) in all combinations, totalling 12 treatments. There were five replicate sea urchins for each treatment (total 60 *T. gratilla*; and total 60 *H. erythrogramma*). At the beginning of the experiments, sea urchins were haphazardly assigned to treatments and measured to obtain initial values for TD (test diameter) and WW (wet weight). TD was the mean of two perpendicular measurements of the aboral side of the test of each sea urchin using a calliper. To obtain initial WW, sea urchins were weighed after removal of excess seawater by holding them out of water for 30 s. There was no significant difference in the initial WW (two-way ANOVA, $p > .05$; mean = 182.4 g ± 6.3 SE) or TD ($p > .05$; mean = 74.5 mm ± 0.8 SE) of *T. gratilla* among treatments. There was also no significant difference in the initial WW ($p > .05$; mean = 111.8 g ± 6.7 SE) or TD ($p > .05$; mean = 64.4 mm ± 1.4 SE) of *H. erythrogramma* among treatments. After they were weighed, sea urchins were placed individually into food grade cylindrical polypropylene containers (11.5 cm ∅ × 8.5 cm high, sealed with matching polypropylene lids). Containers were then placed inside temperature-controlled cabinets for the duration of their allocated air exposure treatment (6, 12, 24, or 48 h). Air temperatures were recorded every 30 min using temperature loggers (Thermodata IBCod 22L): *T. gratilla* (ambient = 27.9 °C ± 2.4 SD; cooled = 11.8 °C ± 1.0 SD; chilled = 3.8 °C ± 0.8 SD); *H. erythrogramma* (ambient = 26.5 °C ± 1.6 SD; cooled = 12.0 °C ± 0.8 SD; chilled = 3.0 °C ± 0.8 SD).

At the end of their allocated air exposure treatment, each sea urchin was reweighed. The change in the weight by sea urchins, hereafter called mass loss (%), was calculated by dividing the final WW by the initial WW multiplied by 100. After their final WW was measured, sea urchins were immediately placed into food grade cylindrical polypropylene containers (12.5 cm ∅ × 9.5 cm high) holding 1.0 L seawater. Containers with living urchins were then placed in a flow-through seawater system where each container was supplied with seawater via a dripper at a rate of 6.0 L h^{-1}. Temperature and salinity of seawater in five haphazardly-selected containers were checked daily using a Hach HQ40d meter with Hach CDC401 conductivity probe (mean temperature and salinity: *T. gratilla* = 25.4 °C ± 0.2 SD, 37.1 ± 0.6 SD; *H. erythrogramma* = 19.8 °C ± 0.4 SD, 34.3 ± 0.8 SD).

After they were returned to water, sea urchins were closely monitored for 3 h and data recorded about whether they had spawned (i.e. released gametes) while out of water, or spawned when returned to water, and whether they floated. Mortality of *T. gratilla* and *H. erythrogramma* was also assessed at the end of the air exposure period and followed for 15 days thereafter. Sea urchins were classed as dead if they failed to display movement, if their spines drooped, and if they were unable to right themselves within 900 s, as per Lares and McClintock (1991). Total mortality was calculated as a percentage for each treatment; the number of living animals divided by the initial number of replicate sea urchins for each treatment, multiplied by 100.

Righting activity was assessed at the beginning of the experiments, 3 h after sea urchins were returned to the water, and then once every 4 days until day 15 following standard protocols (Brothers and McClintock, 2015; Challener and McClintock, 2013). Righting activity was reported as righting success (ability to right within 900 s) and righting time (elapsed time for sea urchins to reorientate themselves once inverted). Sea urchins that were alive but could not reorientate themselves after 900 s were assigned a time of 900 s.

2.4. Effects of type of exposure and drainage on T. gratilla

Mortality and fitness of sea urchins in the air exposure duration × air temperature experiments may have been influenced by the conditions in the sealed containers. For instance, it was observed that any fluid excreted by the sea urchins pooled under their tests, and this fluid may have influenced humidity or inhibited desiccation. To understand whether the conditions that sea urchins were held in while out of water influenced their mortality, wet weight, and righting activity, 40 *T. gratilla* were haphazardly assigned to one of four packaging treatments; all combinations of type of exposure (open vs closed containers) and drainage (drained vs not drained containers). Each treatment had 10 replicate sea urchins.

At the beginning of the experiment, initial WW and TD of *T. gratilla* were measured as previously described (section 2.3). Sea urchins (mean WW = 217.5 g ± 11.3 SE, mean TD = 80.5 mm ± 1.3 SE) were placed individually into food grade cylindrical polypropylene containers (11.5 cm ∅ × 8.5 cm high) that were sealed with a matching polypropylene lid (sealed treatment) or left open without a lid (open treatment). The ‘drained’ container treatments housed *T. gratilla* in containers modified by ~75 × 2 mm slots cut into the bottom, allowing any fluid released from the sea urchins to drain into another container nested below. Containers for the 'not drained' treatments were not modified; the sea urchins thus sat in any fluid they excreted. Containers were placed in a temperature-controlled cabinet for 12 h with an air temperature of 28.6 °C ± 0.9 SD (Thermodata IBCod 22L), after which *T. gratilla* were re-weighed and returned to seawater in food grade polypropylene containers (12.5 cm ∅ × 9.5 cm high, holding 1.0 L seawater) in the previously described flow-through seawater system (mean temperature and salinity: 25.2 °C ± 1.4 SD, 35.3 ± 0.3 SD). Mass loss (%), occurrence of spawning, occurrence of floating, mortality, and righting activity were measured as previously described (section 2.3).

2.5. Statistical analysis

Data on mass loss (%) and righting time for *T. gratilla* and *H. erythrogramma* were analysed by two-way permutational analysis of variance (ANOVA) in PRIMER v6 (Anderson, 2001). Two-way ANOVAs were conducted using 'air temperature' and 'duration of air exposure', or 'type of exposure' and 'drainage' as fixed factors, and individual sea urchins as the level of replication. For *H. erythrogramma* righting time data, the 48 h duration of air exposure treatments were not analysed because of high mortality in the ambient air temperature treatment (Appendix A; Fig. A2). Initial WW or TD were included as covariates in the models for change in weight and righting time respectively but removed when $p > .25$ (Quinn and Keough, 2002). Assumptions of normality and heterogeneity of variance were examined using Q-Q residual plots, values for skewness and kurtosis, and Kolmogorov-Smirnov and Shapiro-Wilk tests, using IBM SPSS v28 (Field, 2013; Quinn and Keough, 2002). All data were not normally distributed, except for righting time in both *T. gratilla* experiments, and were square-root transformed to normalise the data. As analyses using transformed data gave the same outcomes as untransformed data, the results of analyses of untransformed data were presented. Box plots were used to identify outliers. For all datasets, 2–4 outliers were identified. The outcomes of analyses done with outliers included were the same as with outliers excluded, so results from analyses with all data included were presented. For ANOVA analyses, pair-wise comparisons of untransformed data were generated using Euclidean distance and 9999 permutations of the raw data. Post-hoc pair-wise tests were performed when ANOVA results indicated that there were significant differences within treatments with three levels. Monte Carlo values were used if the number of permutations was low (<100). For reporting post-hoc tests, > and < indicate significantly greater than or less than at $p < .05$ respectively, while = indicates no significant difference, $p > .05$.

To assess the potential for air temperature and duration of exposure to air to affect mortality of *T. gratilla* and *H. erythrogramma* to 15 days and righting success after 3 h recovery, a binomial value was assigned to each sea urchin: 0 = dead; 1 = alive; and 0 = did not right within 900 s, 1 = righted within 900 s. Likewise, to assess the potential for type of exposure and drainage to affect mortality of *T. gratilla* to 15 days and righting success after 3 h recovery, a binomial value was assigned to each sea urchin: 0 = dead; 1 = alive; and 0 = did not right within 900 s, 1 = righted within 900 s. Binomial data were analysed using binary logistic regressions in IBM SPSS v28 software. Mortality and righting success were tested against the predictor variables ‘air temperature' (categorical), 'duration of air exposure' (categorical), or 'type of exposure' (categorical), 'drainage' (categorical), and mass loss (%), initial WW, TD, occurrence of spawning, and occurrence of floating after return to water (Appendix A; Fig. A3). Model χ^2 tests were used to select the best-fitting model out of models containing all possible combinations of predictors and their interactions. Wald χ^2 tests were used to assess significant effects of predictors ($p < .05$), and 95% bias-corrected and accelerated confidence intervals (BCa) were calculated based on >1000 bootstrap samples. For *H. erythrogramma* righting success, the 48 h duration of air exposure treatments were not formally analysed due to high mortality in the ambient air temperature treatment (Appendix A; Fig. A2).

Righting times for *T. gratilla* and *H. erythrogramma* measured prior to the air exposure × air temperature experiments were compared with corresponding righting times of all living sea urchins measured during the 15 days after they were returned to water using one-tailed paired samples t-tests in SPSS v28 (Field, 2013).

3. Results

3.1. Temperatures during wet and dry live freight

Temperatures were recorded in separate 20-L polystyrene foam boxes transporting *T. gratilla* in seawater from Australia to Japan (Fig. 1a). Over a 93 h shipping period, similar average temperatures of 25.7 °C ± 3.1 SD and 25.9 °C ± 3.2 SD were recorded in the two boxes (Fig. 1a). The maximum temperature recorded by the loggers was 34.6 °C and the minimum temperature was 21.3 °C (Fig. 1a). Periods of declines in temperatures tended to coincide with times when the boxes were travelling by airplane, while periods of rapidly increasing and maximum temperatures tended to coincide with periods when the boxes were between flights or travelling by road.

Temperatures were also recorded in a 14-L polystyrene box transporting *T. gratilla* dry from Coffs Harbour, Australia to Adelaide, Australia (Fig. 1b). During the 27.5 h shipping period, the average temperature was 21.7 °C ± 1.2 SD (Fig. 1b). The maximum temperature recorded by the logger was 25.5 °C and the minimum temperature was 18.5 °C (Fig. 1b).

3.2. Effects of air temperature and duration of air exposure on mortality

Mortality of adult sea urchins, *T. gratilla* and *H. erythrogramma*, kept out of water for 6–48 h in three air temperature treatments and then returned to seawater for 15 days varied from 0–100% across treatments (Fig. 2). Most sea urchins survived the period that they were kept out of water, and almost all mortality occurred within the first 7 days after the sea urchins were returned to the water (Appendix A; Fig. A1, Fig. A2).

A binary logistic regression analysis testing the effects of air temperature and duration of air exposure treatments on mortality of *T. gratilla* to 15 days was statistically significant χ^2 (3) = 57.96, $p < .00001$ (Table 1; Fig. 2). The model explained 83% (Nagelkerke R^2) of the variation in mortality of *T. gratilla* and correctly classified 91.7% of cases (Table 1). Air temperature was a significant predictor of mortality (Table 1; Fig. 2). Individuals in the cooled treatment (~12 °C) were ~53 times less likely to die than those in the chilled treatment (~4 °C) (Table 1). The likelihood of mortality was not significantly different for *T. gratilla* held in the chilled (~4 °C) or ambient (~27 °C) treatments (Table 1). The percentage of mass lost by *T. gratilla* was also a significant predictor of mortality (Table 1; also see Fig. 3). For every one unit increase in the percentage of mass lost by a sea urchin, the likelihood of mortality increased by 1.63 times (Table 1). Inclusion of other predictors, duration of air exposure, occurrence of spawning, occurrence of floating, initial WW or TD, did not improve the model.

A binary logistic regression analysis testing the effects of air temperature and duration of air exposure on mortality of *H. erythrogramma* to 15 days following exposure was statistically significant χ^2 (5) = 57.05, $p < .00001$ (Table 1; Fig. 2). The model explained 82% (Nagelkerke R^2) of the variation in mortality of *H. erythrogramma* and correctly classified 86.7% of cases (Table 1). The percentage of mass lost by *H. erythrogramma* was the only significant predictor of mortality (Table 1; also see Fig. 3). For every one unit increase in the percentage of mass lost by a sea urchin, the likelihood of mortality increased by 1.81 times (Table 1). Air temperature was included as a predictor in the best-fitting model but was not a significant predictor of mortality (Table 1). Inclusion of other predictors, duration of air exposure, occurrence of spawning, occurrence of floating, initial WW, or TD, did not improve the model.

3.3. Effects of air temperature and duration of air exposure on wet weight

Sea urchins, *T. gratilla* and *H. erythrogramma*, lost 5–53% of their initial WW when kept out of water for up to 48 h in three air temperature treatments (Fig. 3). The percentage of mass lost by *T. gratilla* depended on a significant interaction between air temperature and duration of air exposure (Fig. 3, Table 2). For *T. gratilla* kept out of water for 6 h, there was no statistical difference in the percentage of mass lost by individuals in different air temperature treatments (Fig. 3; Table 2). For *T. gratilla* kept out of water for 12 h, the percentage of mass lost by individuals was 1.2 times greater in the ambient treatment (~27 °C) than the chilled treatment (~4 °C), but neither was significantly different to the cooled treatment (~12 °C) (Fig. 3; Table 2). For *T. gratilla* kept out of water for 24 and 48 h, the percentage of mass lost by sea urchins in the ambient treatment was 2–3 times greater than for those in the chilled and cooled treatments, respectively, and there was no difference in the percentage of mass lost by *T. gratilla* in the chilled and cooled treatments (Fig. 3; Table 2).

The percentage of mass lost by *H. erythrogramma* depended on a significant interaction between air temperature and duration of air exposure (Fig. 3; Table 2). For *H. erythrogramma* kept out of water for 6 h, there was no statistical difference in the percentage of mass lost by sea urchins in different air temperature treatments (Fig. 3; Table 2). For *H. erythrogramma* kept out of water for 12 h, the percentage of mass lost by individuals in the ambient treatment (~27 °C) was 2 times greater than for those in the chilled (~4 °C) and cooled (~12 °C) treatments (Fig. 3; Table 2). For *H. erythrogramma* kept out of water for 24 and 48 h, the percentage of mass lost by sea urchins in the ambient treatment was 1.2 and 3 times greater than for those in the cooled and chilled treatments respectively, and the percentage of mass lost by *H. erythrogramma* in the cooled treatment was 2 times greater than for those in the chilled treatment (Fig. 3; Table 2).

3.4. Effects of air temperature and duration of air exposure on righting activity

Sea urchins, *T. gratilla* and *H. erythrogramma*, varied in their ability to right themselves within 900 s after they were kept out of water for 6–48 h at three air temperatures and then returned to seawater at 25 or 20 °C respectively for 3 h (Fig. 4a). All *T. gratilla* and *H. erythrogramma* kept out of water for 6 h righted themselves within 900 s, whereas most sea urchins kept out of water for 48 h were unable to right themselves (Fig. 4a).

A binary logistic regression analysis testing the effects of air temperature and duration of air exposure on righting success of *T. gratilla* was statistically significant χ^2 (5) = 38.78, p < .001 (Table 1). The best-fitting model explained 66% (Nagelkerke R^2) of the variation in righting success among *T. gratilla* and correctly classified 91.4% of cases (Table 1). Temperature was a significant predictor of righting success (Table 1). *T. gratilla* in the cooled treatment (~12 °C) were ~9.7 times more likely to right themselves than those in the ambient treatment (~27 °C), though this should be interpreted with caution as the 95% CI (BCa confidence interval) for b crossed zero (Table 1). Righting success was not significantly different between *T. gratilla* in the chilled (~4 °C) and ambient treatments (Table 1). Duration of air exposure was included in the best-fitting model, but was not a significant predictor of righting success (Table 1). Inclusion of other predictors, occurrence of spawning, occurrence of floating, the percentage of mass lost while out of water, initial WW or TD, did not improve the model.

A binary logistic regression analysis was used to test the effects of air temperature and duration of air exposure on righting success of *H. erythrogramma* excluding 48 h treatments because of high mortality in the ambient air temperature treatment (Appendix A; Fig. A2). There were no significant predictors among the factors examined, which included air temperature, duration of air exposure, occurrence of spawning, occurrence of floating, initial WW, and TD.

There was substantial variation in the amount of time taken by sea urchins, *T. gratilla* and *H. erythrogramma*, to right themselves after they were kept out of water for 6–48 h at three temperatures and then returned to seawater for 3 h (Fig. 4b). All *T. gratilla* and *H. erythrogramma* kept out of water for 6 h righted themselves in less than 130 s, whereas sea urchins kept out of water for 12–48 h generally took more than 100 s to right themselves (Fig. 4b) or were unable to right themselves and given a score of 900 s (Fig. 4a; Fig. 4b).

The time taken by *T. gratilla* to right themselves depended on a significant interaction between air temperature and duration of air exposure (Fig. 4b; Table 1). For *T. gratilla* kept out of water for 6 or 48 h, there was no difference in amount of time taken by individuals in different air temperature treatments to right (Fig. 4b; Table 1). For *T. gratilla* kept out of water for 12 h, individuals in the chilled treatment (~4 °C) took 900 s to right, and were 4–15 times slower to right than sea urchins in the cooled (~12 °C) and ambient (~27 °C) treatments (Fig. 4b; Table 2). The amount of time to right for *T. gratilla* kept out of water for 12 h in the cooled and ambient treatments was not statistically different (Table 2). For *T. gratilla* kept out of water for 24 h, individuals in the cooled treatment righted at least 4–5 times quicker than those in the chilled or ambient treatments (Fig. 4b; Table 2). The amount of time to right for *T. gratilla* kept out of water for 24 h in cooled and ambient treatments was not statistically different (Table 2).

Being kept out of water appeared to have lasting effects on the amount of time taken by *T. gratilla* to right. One-tailed paired samples t-tests indicated that *T. gratilla* that were alive 13 days after they were kept out of water took significantly longer to right than they did at the beginning of the experiment (initial mean righting time 69 s ± 12 SE; day 13 mean righting time 127 s ± 30 SE; $t(31) = -1.97$, $p = .029$).

The time taken by *H. erythrogramma* to right themselves depended on a significant interaction between air temperature and duration of air exposure (Fig. 4b; Table 2). For *H.*

erythrogramma kept out of water for 6 or 12 h, similar times were taken by individuals in all temperature treatments to right themselves (Fig. 4b; Table 2). For *H. erythrogramma* kept out of water for 24 h, individuals held at ~4 °C took significantly longer to right themselves than at ~12 °C (Fig. 4b; Table 2), but neither treatment were significantly different to ~27 °C. Data from 48 h treatments were not formally analysed due to high mortality in the ambient air temperature treatment (Fig. 4b; Appendix A; Fig. A2), though all *H. erythrogramma* exposed to air for 48 h in the chilled treatments did not right (Fig. 4a) and were assigned a time of 900 s, whereas those in the cooled treatment righted in an average of 266 s (Fig. 4b).

Being kept out of water appeared to have lasting effects on the amount of time taken by *H. erythrogramma* to right. One-tailed paired samples t-tests indicated that *H. erythrogramma* that were alive 8 days after they were kept out of water took longer to right than they did at the beginning of the experiment (initial mean righting time 69 s ± 6 SE; day 8 mean righting time 83 s ± 6 SE; t (31) = -1.73, p = .047), but at 13 days post-exposure, righted in a similar amount of time as they did at the beginning of the experiment (day 13 mean righting time 76 s ± 6 SE; t (30) = -0.69, p = .249).

3.5. Effects of type of exposure and drainage on T. gratilla

Adult *T. gratilla* kept out of water for 12 h at ~28 °C had similar responses regardless of whether they were held in enclosed or open containers or whether any fluid they lost was drained away or allowed to pool around their test (Fig. 5). Total mortality to 15 days post-exposure ranged from 10–40% (Fig. 5a). Using a binary logistic regression analysis the only statistically significant model (χ^2 (1) = 7.14, p < .008) included one predictor – mass loss (%), and explained 24% (Nagelkerke R^2) of the variation in mortality among *T. gratilla* at day 15 and correctly classified 72.5% of cases (Table 3). For every one unit increase in the percentage of mass lost by a sea urchin, the likelihood of mortality increased by ~1.2 times (Table 3). Inclusion of other predictors, type of exposure, drainage, occurrence of spawning, occurrence of floating, initial WW, or TD, did not improve the model.

The percentage of mass lost by *T. gratilla* varied between 28–40% (Fig. 5b), but was not influenced by type of exposure, drainage, or an interaction between these factors (Table 4).

Most *T. gratilla* were able to right themselves after the 12 h air exposure and a 3 h recovery period (90–100%, Fig. 5c). A binary logistic regression analysis was used to test the effects of type of exposure and drainage on righting success, but there were no significant predictors among the factors examined, which included type of exposure, drainage, occurrence of spawning, occurrence of floating, initial WW, and TD.

The amount of time taken by *T. gratilla* to right varied between 160–224 s (Fig. 5d), but was not influenced by type of exposure, drainage, or an interaction between these factors (Table 4).

4. Discussion

Shipping live sea urchins to market is a key part of the aquaculture production process that is often overlooked by researchers. In this study, it was found that two sea urchin species, *Tripneustes gratilla* and *Heliocidaris erythrogramma*, can be held out of water for up to 24 h with low mortalities of between 0–20% if cooled to ~12 °C. Higher (~27 °C) or lower temperatures (~4 °C) caused increased mortality and impacted the health of the sea urchins. Increasing exposure times for sea urchins from 24 h at 12 °C to 48 h increased mortality to 60% for the subtidal *T. gratilla* but mortality remained at 20% for the intertidal *H. erythrogramma*. After a 24 h exposure period, mortality reported here for *T. gratilla* was less than that previously reported for another subtidal urchin, *Mesocentrotus franciscanus* (Warren and Pearce, 2020). Survival of subtidal/intertidal *H. erythrogramma* following 24 and 48 h of air exposure was similar to another sea urchin *Strongylocentrotus purpuratus* that is also found in the low intertidal (James and Evensen, 2022). For both *T. gratilla* and *H. erythrogramma*, the amount of fluid lost during emersion was a strong predictor of mortality, indicating that investigating ways of reducing the loss of fluid by sea urchins when emersed from water might be a good way to extend dry shipping times.

Mortality of sea urchins when held out of water was correlated with mass lost caused mostly from fluid being released from the mouth of the sea urchins. With every 1% increase in mass loss, mortality rose by 1.63 and 1.81 times for *T. gratilla* and *H. erythrogramma*, respectively. A sea urchin’s tube feet collapse when coelomic fluid is lost and it is the tube feet that sea urchins rely on for gas exchange (Fenner, 1973; Johansen and Vadas, 1967). It is likely then that collapse of tube feet led to mortality by limiting the animals' capacity for gas exchange in air and during initial re-immersion in water. Counterintuitively, for *S. purpuratus* very small losses of coelomic fluid during emersion (0.07% compared to >5% loss found in this study) creates an air-water boundary in the oesophagus enabling this species to oxygenate its celomic fluid and continue to respire despite having collapsed tube feet (Burnett et al., 2002; Fenner, 1973). Spicer et al. (1998) found that sea urchins *Psammechinus miliaris* and *Echinus esculentus* compensated for rising CO_2 concentrations in their coelomic fluid by releasing Ca^{2+} and Mg^{2+} ions, preventing acidosis; an option not available to sea urchins exposed to air. Warren and Pearce (2020) theorised that mass loss caused sea urchins to float, reducing survival. Though a substantial proportion of *T. gratilla* and *H. erythrogramma* lost sufficient fluid to cause them to float, we did not find evidence of a relationship between mortality and floating.

Maintaining *T. gratilla* and *H. erythrogramma* at ~12 °C compared to ~4 °C or ~27 °C improved their survival when exposed to air. James and Evensen (2022) reported that *S. droebachiensis* exposed for periods lasting between 12–28 h were 1.6 times more likely to survive when kept at 4 °C compared to 15 °C. The difference between the temperatures that caused the lowest mortality for *S. droebachiensis* (4 °C, James and Evensen, 2022) and *T. gratilla* and *H. erythrogramma* (~12 °C, this study) might reflect differences in thermal tolerances and the habitats where these species are found. *T. gratilla* and *H. erythrogramma* used here were from the tropical to warm temperate east coast of Australia, while *S. purpuratus* occupies cool temperate waters stretching from California to Alaska (Keesing, 2013; Lawrence and Agatsuma, 2013; Rogers-Bennett, 2013). *T. gratilla* regularly experiences the upper temperature used here (~27 °C) in its natural range. *H. erythrogramma* also experience temperatures ≥27 °C in tide pools (e.g., Wolfe et al., 2020), though higher latitude populations might find this temperature stressful (Minuti et al., 2021).

Although they may naturally experience temperatures ≥27 °C, both *T. gratilla* and *H. erythrogramma* suffered increased mortality when emersed at this temperature. This is unsurprising as thermal tolerance can be reduced when combined with other stressors (Sokolova and Pörtner, 2001). During emersion and desiccation, sea urchins likely experience multiple stressful physiological effects simultaneously. For instance, higher temperatures increase basal metabolic rates of ectotherms creating a higher oxygen demand, while exposure causes internal pH and oxygen concentrations to decline, reducing the ability of organisms to satisfy this higher oxygen demand (Cui et al., 2020; Haukenes et al., 2009; Lorenzon et al., 2007; Miller et al., 2002). Concentrations of ammonia, a by-product of protein metabolism, also increase with increases in temperature and this causes a rise in oxidative stress and toxicity, which can prove fatal (Harianto et al., 2021). An inability to exchange fluids during emersion is likely to exacerbate this effect (Lu et al., 2020). It is likely that the chilled treatment (~4 °C) did not increase survival because neither *T. gratilla* or *H. erythrogramma* would experience temperatures this low in the wild. Exposing temperate and tropical sea urchins to temperatures below 8 °C is often stressful and can cause death (e.g., Collin et al., 2018; van der Walt et al., 2021). This is highlighted by the fact that the tropical *T. gratilla* experienced 100% mortality after 12 h exposure at ~4 °C but the more temperate *H. erythrogramma* survived at this temperature for 12 h but succumbed by 24 h. When moved from seawater to the 4 °C and 12 °C air exposure treatments, *T. gratilla* experienced a greater fluctuation in temperature than *H. erythrogramma*. The higher mortality of *T.gratilla* in the 4 °C treatment may result from this larger temperature change. It would be interesting to see if acclimation to shipping temperature reduced mortality rates.

The average mass lost by *T. gratilla* and *H. erythrogramma* during 6 h of emersion was similar, at 8.1% and 5.7% respectively. Burnett et al. (2002) reported that *S. purpuratus* lost 0.07% of its mass during 8 h exposure, much less than what was reported here. Longer periods of aerial exposure and chilled (~4 °C) and ambient shipping temperatures (~27 °C) correlated with increased mass loss by *T. gratilla* and *H. erythrogramma*. This study is the first to report the relationship between temperature and exposure time on mass lost in an echinoderm. Several studies have also found a similar relationship between duration of aerial exposure or temperature on the amount of weight lost in gastropods and bivalves (Delorme et al., 2021; Sokolova and Pörtner, 2001; Williams and Morritt, 1995), but it is likely the mechanism by which water is lost varies among taxa. We observed the majority of mass loss from sea urchins was through direct release of internal fluid via the mouth. Reducing water loss might be the key to extending emersion times for sea urchins but there is little evidence of how best to do this from the present study apart from moderating the temperature.

Righting activity is a good indicator of stress and fitness in echinoderms, with longer righting times correlated with greater stress (Brothers and McClintock, 2015; Challener and McClintock, 2013). In this study, similar to the mortality data, longer periods out of water and warm or cold temperatures increased the time that *T. gratilla* and *H. erythrogramma* took to right themselves when returned to seawater. Echinoderms right themselves the quickest within their thermal limits (Ubaldo et al., 2007). However, in this study, the test of righting behaviour was done in benign temperatures of 20 °C or 25 °C respectively; thus, the carryover effect of stressful temperature and exposure treatments were measured in the righting experiments. Slow righting times for both sea urchin species exposed to air at ~4 °C is unsurprising as these temperatures would normally not be experienced by *T. gratilla* or *H. erythrogramma* and thus should be stressful. However, the ambient air temperature treatment (~27 °C) was within the natural thermal range experienced by these sea urchins, and close to the thermal optima of *T. gratilla* when submerged in water (Mos et al., 2015). As with mortality, thermal tolerances are likely to vary when other stressors are present, and exposure to air at ~27 °C creates a unique set of challenges compared to the same temperatures experienced while submerged. This indicates that thermal tolerances measured in water are likely a poor predictor of which temperatures minimise stress and mortality during dry shipping.

Righting times for *T. gratilla* and *H. erythrogramma* were measured before and after the experimental treatments as a measure of sea urchin health. For *H. erythrogramma*, righting times recovered completely to the pre-test times by 15 days following emersion, indicating that these sea urchins had fully recovered from stress caused by exposure to air in 15 days or less. However, *T. gratilla* did not recover fully from the emersion treatment; even after 15 days of recovery, they did not right themselves as quickly as before the treatments. This indicates that even though there was very low mortality for *T. gratilla* for many of the emersion treatments, there was a subtle decrease in their fitness after they were exposed to air. Whether this is likely to have an impact on the quality of the gonads of the sea urchins is unknown but should be examined.

The gonad condition of either species was not surveyed before the experiments but given the history and holding conditions of the sea urchins before the experiment we would expect the large majority would be reproductively competent. Spawning occurred in 50% of *T. gratilla* exposed to air for 12–48 h. Air exposure can stimulate sea urchins to spawn and, for some species, the likelihood of spawning increases with longer exposure periods (Gago and Luís, 2011; James and Evensen, 2022). Sudden temperature fluctuations can also stimulate spawn spawning (Llodra, 2002). Spawning by *H. erythrogramma* in the experiment occurred in only one animal. There was a correlation between spawning and mortality of *T. gratilla*, but this may not be a major problem for the shipping of farmed sea urchins. Uni (sea urchin gonads) obtain a higher market price before they mature (Phillips et al., 2010). This is because nutrients are stored within the gonads as nutritive phagocytes, and it is these cells that give the gonads their distinctive taste and mouthfeel (Phillips et al., 2010). As the gonads mature, nutritive phagocytes are converted to gametes (Tenuzzo et al., 2012; Dumont et al., 2006) making them watery and bitter, and reducing their market price (Tenuzzo et al., 2012; Dumont et al., 2006). It can be assumed in most cases that sea urchins will be shipped while in a pre-spawn condition, but even if it becomes a problem it is possible to reduce spawning in transport by injection with Mg_2Cl (Arafa et al., 2007; Arafa et al., 2010). In future research, gonad quality and stage should be measured prior to and after urchins are exposed to air to ensure they are still marketable. Sea urchins are also shipped live for the ornamental aquarium trade and for use in aquaculture as broodstock or for conditioning. Urchin spawning is not an issue for these markets.

A final experiment examined whether packing *T. gratilla* in drained containers and closed containers affected fitness and survival of urchins during 12 h of emersion. This was done to test whether fluid released from the sea urchins that pooled in the bottom of their containers or the increased humidity of closed containers benefitted *T. gratilla* exposed to air. The type of packing had no impact with fitness and mortality similar among all treatments. Similar to our previous experiments, the proportional mass lost by *T. gratilla* was a significant predictor for mortality. For every 1% increase in mass loss, the likelihood of mortality rose by 1.2 times. It is common for industry to use semi-dry methods that involve packaging sea urchins alongside seawater-soaked hessian sacks when being transported out of water (James and Evensen, 2018; Warren and Pearce, 2020). This study found no clear benefit of partially immersing *T. gratilla* in fluid that sea urchins released throughout the 12 h exposure period on its mortality and fitness. These results suggest producers transporting *T. gratilla* up to 12 h can reduce costs by packing *T. gratilla* using dry rather than semi-dry methods while not compromising on the fitness and health of the sea urchins upon arrival. All treatments showed high survival and fitness among *T. gratilla*, indicating this species was not inherently stressed by the 12 h exposure period.

5. Conclusions

This study found that live *T. gratilla* and *H. erythrogramma* can be transported to market using dry shipping methods. Mortality was lowest for up to 24 h aerial exposure when sea urchins were kept at ~12 °C. The proportion of mass lost by sea urchins was an important predictor of mortality. Future research focusing on reducing internal water loss during emersion might be a way to extend dry shipping times.

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Credit authorship contribution statement

Joshua Bennett: Methodology, Investigation, Data Curation, Writing - Original Draft **Benjamin Mos:** Conceptualisation, Methodology, Formal Analysis, Resources, Data Curation, Writing – Original Draft, Writing – Review and Editing, Visualisation, Supervision **Symon Dworjanyn:** Conceptualisation, Methodology, Resources, Writing – Original Draft, Writing – Review and Editing, Supervision

Data availability

Data is available on reasonable request.

Declaration of Interests

Benjamin Mos reports financial support was provided by the Australian Research Council. Symon A. Dworjanyn reports membership of the board of Mugenuni Pty Ltd, which had no role in the project's conception, design, outcomes, or the decision to publish the study. Joshua Bennett has no conflicts of interest to declare.

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Figures

Fig. 1. Temperatures experienced by live sea urchins during road and air freight. (a) Temperatures recorded by two temperature loggers in boxes transporting *Tripneustes gratilla* from Coffs Harbour, Australia to Ginoza, Okinawa Island, Japan (27–30 July 2017). *T. gratilla* were packed using conventional methods for transporting live marine invertebrates in the live aquarium trade; submerged in seawater with O_2 gas headspace inside plastic fish transport bags, housed in 20-L polystyrene foam boxes. (b) Temperatures recorded by a temperature logger in a package containing *T. gratilla* packed without seawater (i.e. 'dry') in sealed plastic fish transport bags housed in a 14-L polystyrene foam box, and transported from Coffs Harbour, New South Wales, Australia to Adelaide, South Australia, Australia (21–22 October, 2022).

Fig. 2. Total mortality of adult sea urchins, *Tripneustes gratilla* and *Heliocidaris erythrogramma*, held out of water for 6–48 h in three air temperature treatments (chilled ~4 °C; cooled ~12 °C; ambient ~27 °C) and then returned to seawater for 15 days (also see Appendix A; Fig. A1; Fig. A2). Data expressed as a percentage of sea urchins which died out of the total number of sea urchins in each treatment (n = 5). 0 indicates treatments where there was no mortality.

Fig. 3. The percentage of body mass lost by adult sea urchins, *Tripneustes gratilla* and *Heliocidaris erythrogramma*, when kept out of water for 6–48 h in three air temperature treatments (chilled ~4 °C; cooled ~12 °C; ambient ~27 °C). Mass loss (%) was calculated for each sea urchin as the difference between the initial and final WW (wet weight), divided by the initial WW × 100. Data are means ± SE, n = 5.

Fig. 4. Righting activity of adult sea urchins, *Tripneustes gratilla* and *Heliocidaris erythrogramma*, when kept out of water for 6–48 h in three air temperature treatments (ambient ~27 °C; cooled ~12 °C; chilled ~4 °C) and then returned to seawater for 3 h. (a) The percentage of sea urchins which righted themselves within 900 s. 0 indicates treatments where no sea urchin righted within 900 s. Data for *H. erythrogramma*/48 h/ambient not available due to high mortality in this treatment (Appendix A; Fig. A2). Data are means ± SE, n = 5 except *T. gratilla* ambient/48 h treatment where n = 3. (b) Elapsed time for sea urchins to right themselves. Sea urchins that were alive, but did not right by 900 s were allocated a time of 900 s. Data for *H. erythrogramma*/48 h/ambient not available due to high mortality in this treatment (Appendix A; Fig. A2). Data are means ± SE, n = 5 except *T. gratilla* ambient/48 h treatment where n = 3. NA = data not available.

Fig. 5. Nil effects of type of exposure and drainage on sea urchin *Tripneustes gratilla* held out of water at a mean air temperature of 28.6 °C ± 0.9 SD for 12 h. (a) Total mortality after 15 days. (b) The amount of mass lost by *T. gratilla* while exposed to air, expressed as a percentage of initial WW (wet weight). (c) Proportion of *T. gratilla* that were able to right themselves within 900 s, measured 3 h after they were returned to seawater. (d) Elapsed time to right, measured 3 h after *T. gratilla* had been returned to seawater. *T. gratilla* were held out of water in containers with a sealed lid (sealed) or without a lid (open), and any fluid released by the sea urchins was drained away (drained) or allowed to pool around the bottom of the sea urchins' tests (not drained). (a,c) Data expressed as a percentage of *T. gratilla* which (a) died or (c) righted within 900 s out of the total number of sea urchins in each treatment (N = 10). (b,d) Data are means ± SE, n = 10 except for righting time/sealed/not drained where n = 9.

Tables

Table 1. Coefficients of models predicting (a,c) mortality and (b) righting success after 3 h recovery of adult sea urchins *Tripneustes gratilla* and *Heliocidaris erythrogramma* assigned to 12 treatments resulting from the factorial cross of three air temperature treatments (chilled ~4 °C; cooled ~12 °C; ambient ~27 °C) and four duration of air exposure treatments (6, 12, 24, 48 h).

Species	Variable	Predictor	Comparison	Wald χ^2	*p*	*b*	Lower	Odds	Upper
T. gratilla	**(a) Mortality**	**Air temperature**		**10.74**	**.005**				
			chilled vs cooled	**9.62**	**.002**	**-3.98 (-21.61, -2.46)**	**0.002**	**0.019**	**0.232**
			chilled vs ambient	3.51	.061				
		Mass loss (%)		**5.61**	**.018**	**0.49 (0.26, 32.45)**	**1.09**	**1.63**	**2.45**
		Constant		3.80	.051				
	(b) Righting success	**Air temperature**		**11.01**	**.004**				
			ambient vs chilled	2.03	.154				
			ambient vs cooled	**5.34**	**.021**	**2.28 (-0.49, 40.02)**	**1.41**	**9.73**	**67.09**
		Duration of air exposure		3.62	.305				
		Constant		**<0.01**	**.998**				
H. erythrogramma	**(c) Mortality**	Air temperature		2.59	.274				
		Mass loss (%)		**7.24**	**.007**	**0.59 (0.26, 233.59)**	**1.17**	**1.81**	**2.78**
		Constant		**6.68**	**.010**				

Confidence intervals (95% bias corrected and accelerated) are based on >1000 bootstrap samples. NA, could not be calculated. Bold type indicates predictors with significant differences among levels according to Wald χ^2 tests ($p < .05$). Note (a) $R^2 = .70$ (Homer & Lemeshow), .62 (Cox & Snell), .83 (Nagelkerke). Model χ^2 (3) = 57.96, $p < .00001$. Overall percentage correct = 91.7%. (b) $R^2 = .48$ (Homer & Lemeshow), .49 (Cox & Snell), .66 (Nagelkerke). Model χ^2 (5) = 38.78, $p < .001$. Overall percentage correct = 91.4%. (c) $R^2 = .69$ (Homer & Lemeshow), .61 (Cox & Snell), .82 (Nagelkerke). Model χ^2 (3) =57.05, $p < .00001$.

Table 2. Outcomes of ANOVA analyses examining the effects of three air temperature treatments (chilled ~4 °C; cooled ~12 °C; ambient ~27 °C) and four duration of air exposure treatments (6, 12, 24, 48 h) on the mass loss (%) and righting time of adult sea urchins *Tripneustes gratilla* and *Heliocidaris erythrogramma* in laboratory experiments. df, degrees of freedom; MS, mean square; Temp, air temperature; Expo, duration of air exposure. Significant factors are in bold ($p < .05$). For *H. erythrogramma* righting time, 48 h duration of air exposure treatments were not included in the analysis due to high mortality in the ambient air temperature treatment (Fig. 4b; Appendix A; Fig. A2).

Parameters	**Source**	**df**	**MS**	**F**	***p***	**Post hoc tests**	
T. gratilla						*Interaction*	*Interaction*
Mass loss (%)	**Temp**	**2**	**1.45E3**	**31.12**	**.0001**	chilled (6=12 < 24=48 h)	6 h (chilled=cooled=ambient)
	Expo	**3**	**1.87E3**	**40.22**	**.0001**	cooled (6 < 12=24=48 h)	12 h (chilled=cooled < cooled=ambient)
	Expo × Temp	**6**	**396.85**	**8.54**	**.0001**	ambient (6 < 12 < 24 < 48 h)	24, 48 h (chilled=cooled < ambient)
	Residual	48	**46.49**				
Righting time	**Test diameter**	**1**	**2.51E5**	**5.46**	**.0261**	*Interaction*	*Interaction*
	Temp	**2**	**6.76E5**	**14.68**	**.0001**	chilled (6 < 12=24=48 h)	6, 48 h (chilled=cooled=ambient)
	Expo	**3**	**1.13E6**	**24.60**	**.0001**	cooled (6=12=24 < 12=24=48 h)	12 h (chilled < cooled=ambient)
	Expo × Temp	**6**	**3.76E5**	**8.17**	**.0001**	ambient (6=12 < 24=48 h)	24 h (cooled < chilled=ambient)
	Residual	45	4.60E4				
H. erythrogramma							
Mass loss (%)	Initial wet weight	1	251.94	3.06	.087	*Interaction*	*Interaction*
	Temp	**2**	**2.03E3**	**24.62**	**.0001**	chilled (6=12 < 12=24 < 24=48 h)	6 h (chilled=cooled=ambient)
	Expo	**3**	**1.39E3**	**16.81**	**.0001**	cooled (6=12 < 24=48 h)	12 h (chilled=cooled < ambient)
	Expo × Temp	**6**	**300.14**	**3.64**	**.0034**	ambient (6 < 12 < 24=48 h)	24, 48 h (chilled < cooled < ambient)
	Residual	47	82.42				
Righting time	**Temp**	**2**	**1.82E5**	**4.76**	**.0154**	*Interaction*	*Interaction*
	Expo	**3**	**3.86E5**	**10.13**	**.0007**	chilled (6=12 < 24 h)	6, 12 h (chilled=cooled=ambient)
	Expo × Temp	**4**	**1.71E5**	**4.48**	**.0052**	cooled, ambient (6=12=24 h)	24 h (chilled=ambient > ambient=cooled)
	Residual	**36**	**3.82E4**				

Table 3. Coefficients of model predicting mortality of adult *Tripneustes gratilla* assigned to four type of exposure-drainage treatments.

Predictor	Wald χ^2	*p*	*b*	Lower	Odds	Upper
Mass loss (%)	**4.62**	**.032**	**0.18 (0.03, 0.48)**	**1.02**	**1.20**	**1.41**
Constant	**7.48**	**.006**				

Confidence intervals (95 % bias corrected and accelerated) are based on 1000 bootstrap samples. Bold type indicates predictors with significant differences among levels according to Wald χ^2 tests (p < .05). Note R^2 = .15 (Homer & Lemeshow), .16 (Cox & Snell), .24 (Nagelkerke). Model χ^2 (1) = 7.14, p < .008. Overall percentage correct = 72.5%.

Table 4. Outcomes of ANOVA analyses examining the effects of two type of exposure treatments (open, sealed) and two drainage treatments (drained, not drained) on the mass loss (%) and righting time of adult *Tripneustes gratilla* during a laboratory experiment. df, degrees of freedom; MS, mean square; Expo, type of exposure; Drain, drainage type

Parameters	Source	df	MS	F	*p*
Mass loss (%)	Expo	1	127.31	3.95	.055
	Drain	1	77.56	2.40	.130
	Expo × Drain	1	85.86	2.65	.112
	Residual	36	32.26		
Righting time	Test diameter	1	2.12E5	3.79	.060
	Expo	1	1.89E4	0.34	.565
	Drain	1	1.03E4	0.18	.671
	Expo × Drain	1	5.38E3	0.10	.758
	Residual	34	5.59E4		

Appendix A. Supplementary Data

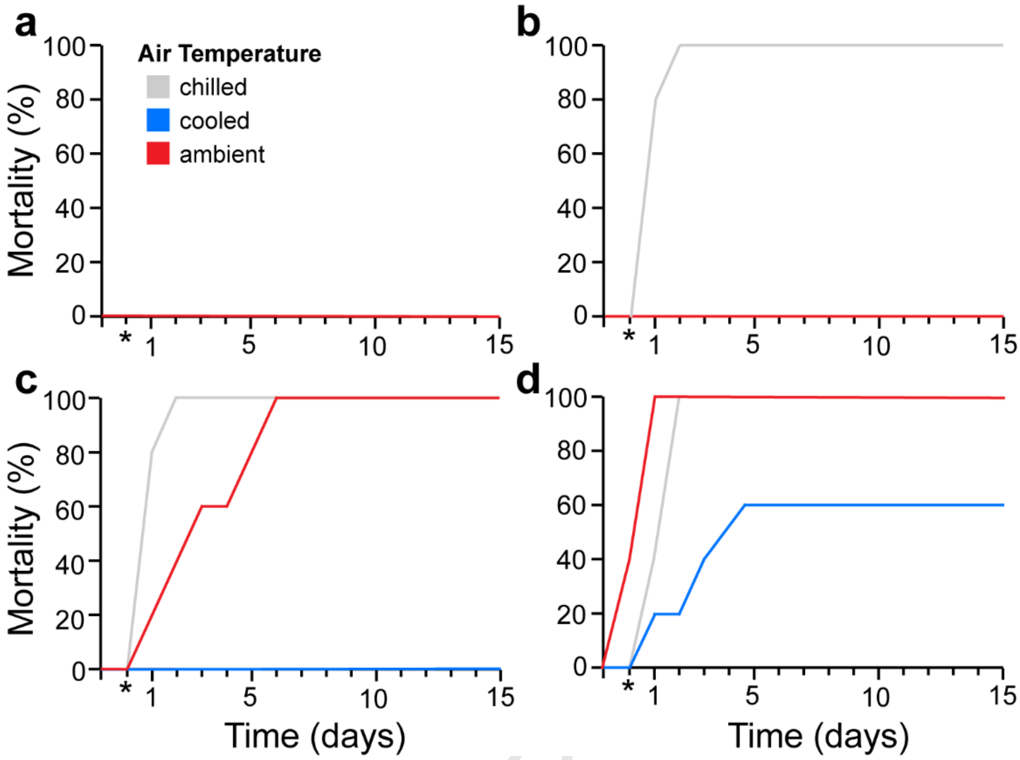


Fig. A1. Cumulative mortality of adult sea urchin, *Tripneustes gratilla*, held out of water for (a) 6 h, (b) 12 h, (c) 24 h, and (d) 48 h in three air temperature treatments (ambient ~27 °C; cooled ~12 °C; chilled ~4 °C) and then returned to seawater (25 °C) for 15 days. Asterisks (*) identify measurements recorded 3 h after sea urchins were returned to the water. Data expressed as a percentage of sea urchins which died out of the total number of sea urchins in each treatment (n = 5).

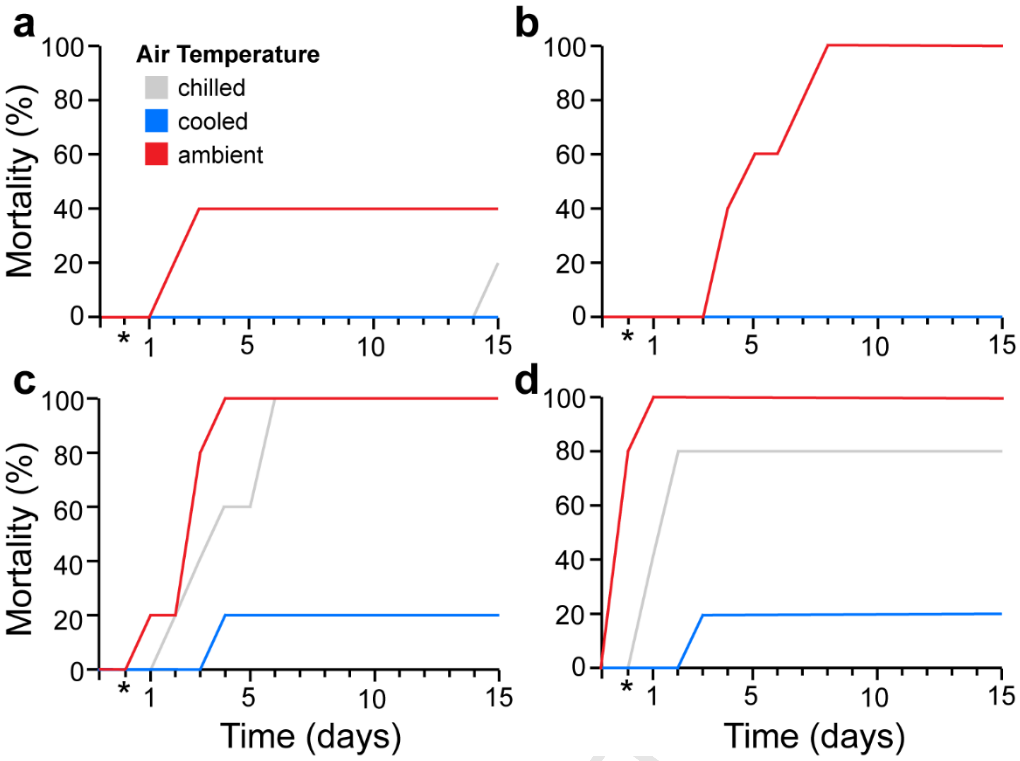


Fig. A2. Mortality of adult sea urchin, *Heliocidaris erythrogramma*, held out of water for (a) 6 h, (b) 12 h, (c) 24 h, and (d) 48 h in three air temperature treatments (ambient ~27 °C; cooled ~12 °C; chilled ~4 °C) and then returned to seawater (20 °C) for 15 days. Asterisks (*) identify measurements recorded 3 h after sea urchins were returned to the water. Data expressed as a percentage of sea urchins which died out of the total number of sea urchins in each treatment (n = 5).

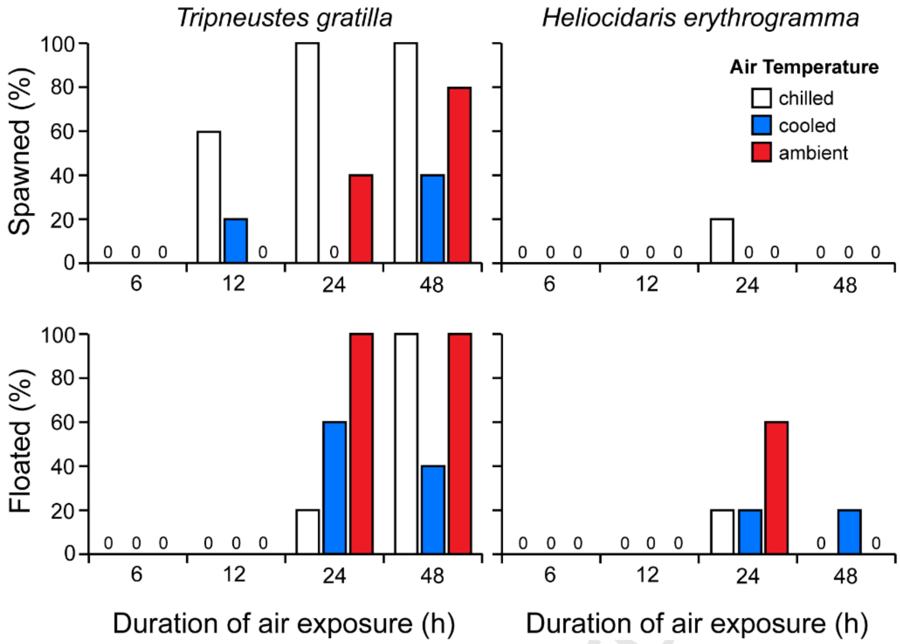


Fig. A3. Responses of adult sea urchins, *Tripneustes gratilla* and *Heliocidaris erythrogramma*, when kept out of water for 6–48 h in three air temperature treatments (ambient ~27 °C; cooled ~12 °C; chilled ~4 °C) and then returned to seawater for 3 h. Data presented as the percentage of sea urchins which (a) spawned while out of water or within 3 h of being returned to water or (b) floated when they were returned to water out of the total number of sea urchins in each treatment (n = 5). 0 indicates treatments where no sea urchin spawned (a) or floated (b).

Authors statement

This study was compliant with all laws and institutional guidelines and did not fall under the criteria for ethics approval.

Declaration of interests

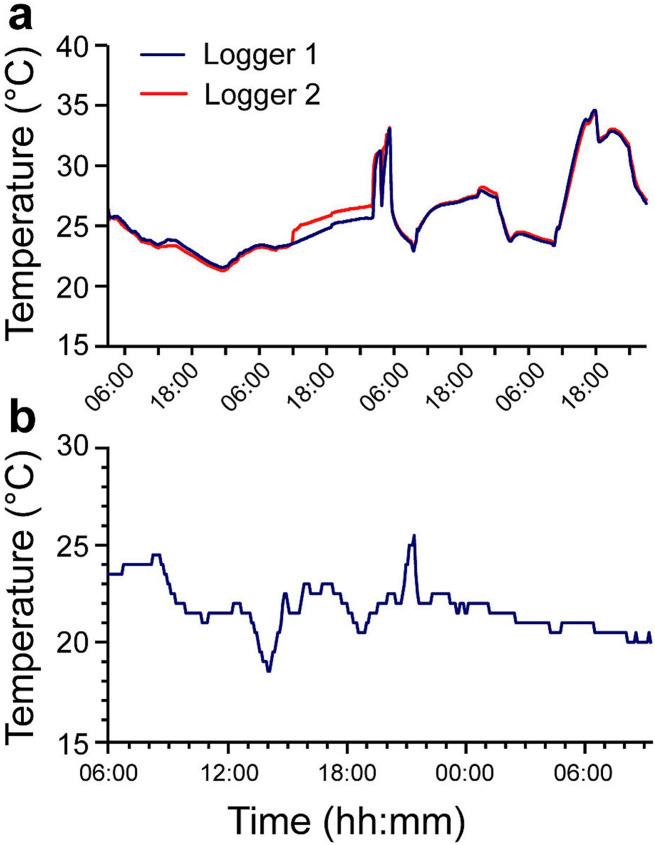
☐The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Benjamin Mos reports financial support was provided by Australian Research Council. Symon Dworjanyn reports a relationship with MUGENUNI Pty Ltd that includes: board membership.

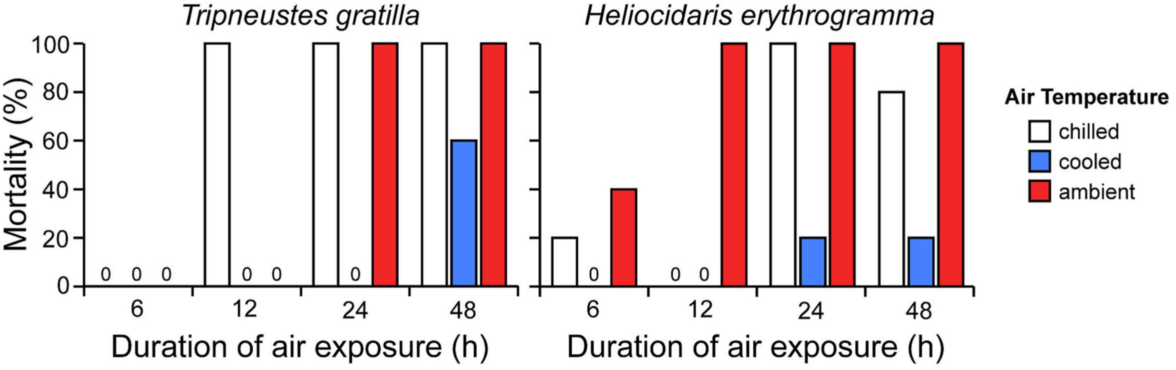
Highlights:

- Sea urchins can be shipped dry, *T. gratilla* (24h) and *H erythrogramma* (48h).
- Maintaining moderate temperatures is important when shipping sea urchins dry.
- Moisture loss correlated with sea urchin mortality.
- Different methods of packing sea urchins dry tested did not improve survival.



a
Logger 1
Logger 2
Temperature (°C)
40
35
30
25
20
15
06:00
18:00
06:00
18:00
06:00
18:00
06:00
18:00
b
Temperature (°C)
30
25
20
15
06:00
12:00
18:00
00:00
06:00
Time (hh:mm)

Figure 1



Tripneustes gratilla
Heliocidaris erythrogramma
Mortality (%)
100
80
60
40
20
0
0 0 0
0 0
0
0
0 0
6
12
24
48
Duration of air exposure (h)
Air Temperature
chilled
cooled
ambient

Figure 2

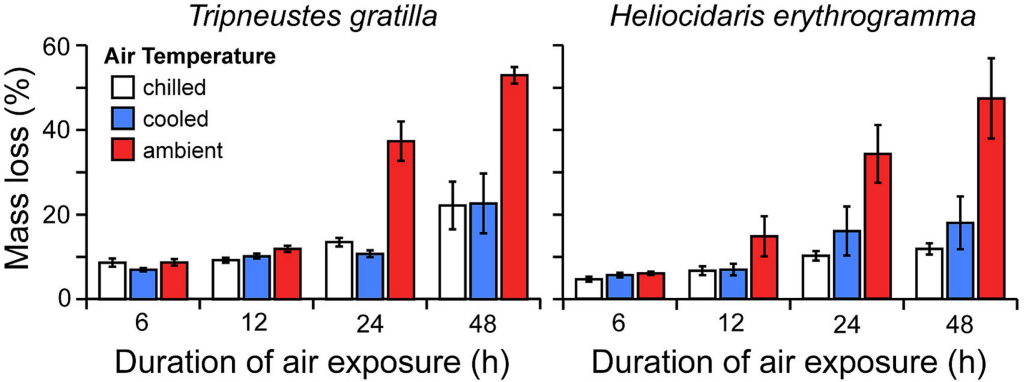

Tripneustes gratilla
Heliocidaris erythrogramma
Air Temperature
chilled
cooled
ambient
Mass loss (%)
60
40
20
0
6
12
24
48
Duration of air exposure (h)
Duration of air exposure (h)

Figure 3

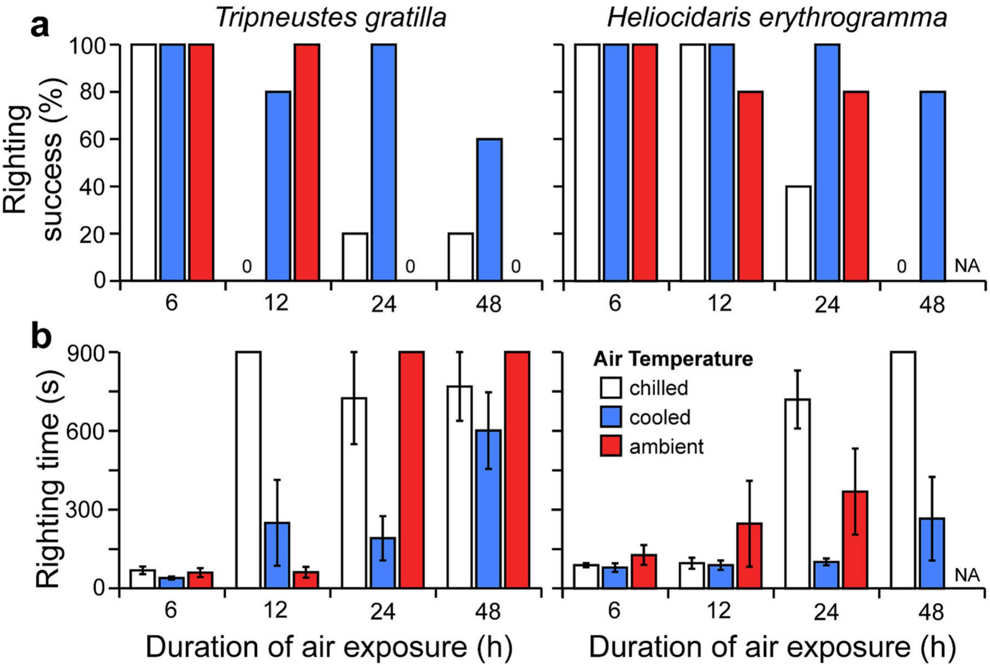

a
Tripneustes gratilla
Heliocidaris erythrogramma
Righting success (%)
100
80
60
40
20
0
0
0
0
0
NA
6
12
24
48
6
12
24
48
b
Righting time (s)
900
600
300
0
Air Temperature
chilled
cooled
ambient
NA
6
12
24
48
6
12
24
48
Duration of air exposure (h)
Duration of air exposure (h)

Figure 4

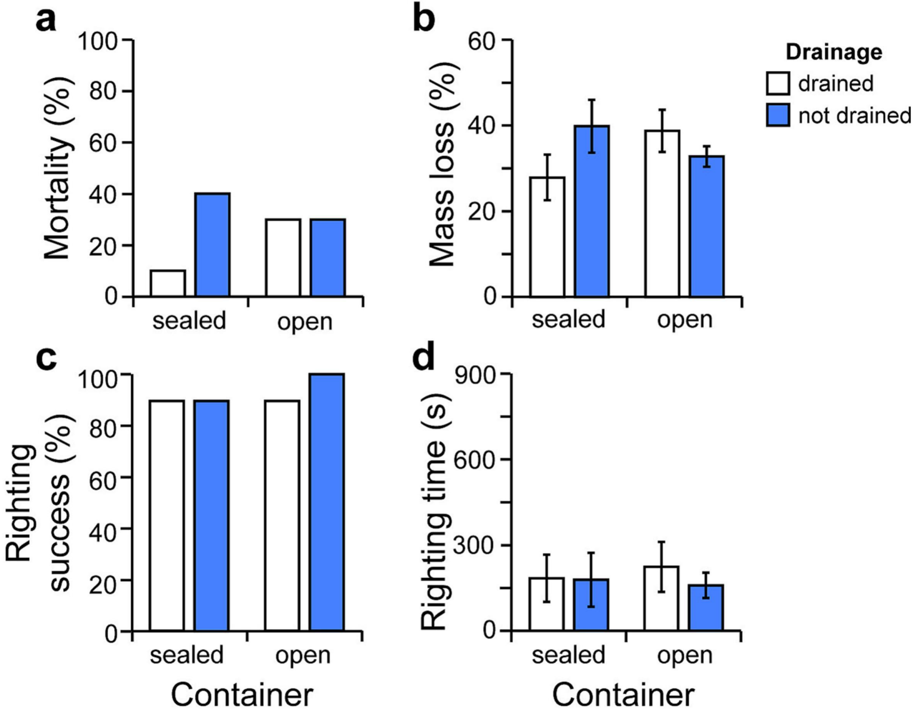

a
b
c
d
Mortality (%)
Mass loss (%)
Righting success (%)
Righting time (s)
100
80
60
40
20
0
900
600
300
sealed
open
Container
Drainage
drained
not drained

Figure 5